# JayPharma Mobile ERP Integration

Main ERP Database Integration & Schema Specification

**API v1.0.2 • Production Ready**

Target System: Microsoft SQL Server
Document Date: November 2026

### 📌 Executive Overview for Main System Engineers

The **JayPharma Mobile Field Collection Application** is used by field medical representatives and collection agents to collect cash/cheque payments from pharmacies and hospital accounts. When an agent confirms a collection on their mobile device, the JayPharma backend service immediately writes the settlement data to the **Main ERP SQL Server Database**.

**Critical Safety Architecture:** The mobile application **NEVER writes to, alters, or locks the `dbo.Invoice` table directly**. Instead, all field collections are inserted as pending audit records into **`dbo.AppInvoice`** and **`dbo.DPay`**. This guarantees 100% data integrity, prevents concurrency collisions with head-office ERP billing users, and provides complete traceability with receipt numbers, collector IDs, and GPS stamps.

## 1. DATA FLOW ARCHITECTURE

```
[Field Agent Mobile App] ──(HTTP JSON / SSL)──► [JayPharma API Gateway] │
┌──────────────────────────────────────────────┘ ▼ ▼ [JayPharmaApp Database]
[Main ERP SQL Server] • Receipts (Header & Metadata) • dbo.AppInvoice (Invoice settlements) • ReceiptInvoiceItems (Allocations)
• dbo.DPay (Daily collection ledger) • AuditLogs (Collector & GPS stamp) (dbo.Invoice remains read-only for field app)
```

## 2. PRIMARY TABLE: `DBO.APPINVOICE` COLUMN DATA DICTIONARY

Every time a receipt is issued, one or more rows are inserted into `dbo.AppInvoice` (one row per allocated invoice, or one row for unallocated advance).

| COLUMN NAME | DATA TYPE | DESCRIPTION & BUSINESS PURPOSE | SOURCE VALUE FROM MOBILE APP | SAMPLE VALUE |
|---|---|---|---|---|
| **`InvNo`** | `INT` | Target Invoice Number being settled. *Set to 0 if this is an Advance payment without invoice.* | Numeric invoice ID from customer dues list, or `0` | `260928001` / `0` |
| **`Date`** | `DATETIME` | Transaction / Receipt date recorded at collection. | Server current date (`DateTime.Now.Date`) | `2026-11-11` |
| **`CCode`** | `INT` | Customer Account Code in Main ERP. | Customer numeric ID code | `65714304` |
| **`CName`** | `NVARCHAR(150)` | Registered Customer / Pharmacy / Hospital name. | Customer name snapshot at time of payment | `B.I.B. Pharmaceuticals` |
| **`InVal`** | `DECIMAL(18,2)` | Total original face value of the invoice. *(0 for Advance)* | `InvoiceAmount` from ERP Invoice | `34444.00` |
| **`SetAmt`** | `DECIMAL(18,2)` | **Amount settled towards this specific invoice** by this receipt. | Allocated collection portion (Rs.) | `1000.00` |
| **`Branch`** | `NVARCHAR(50)` | Company Operating Branch / Division name. | Constant string identifier | `'Jay Pharma'` |
| **`SettleType`** | `NVARCHAR(50)` | Settlement status category:<br>`SETTLED` (Fully cleared, remaining balance ≤ 0.01)<br>`PART SETTLED` (Partial payment, balance remains)<br>`ADVANCE` (Payment on account / unallocated) | Evaluated dynamically based on remaining balance | `'Settled'` / `'Part Settled'` / `'Advance'` |
| **`CrPeriod`** | `INT` | Credit period days placeholder. | Default fixed integer | `0` |
| **`DOther1`** | `DECIMAL(18,2)` | **Remaining Balance** on this invoice after applying `SetAmt`. | `InvoiceAmount - TotalSettledSoFar` | `33444.00` |
| **`DOther2`** | `DECIMAL(18,2)` | **Total Receipt Grand Total** (entire payment amount collected). | Total sum collected across all items | `1000.00` |
| **`Other1`** | `NVARCHAR(50)` | **Manual Receipt Number (MR No)** generated by mobile app. | Collector-scoped receipt number | `'KANDY-026482'` |

| COLUMN NAME | DATA TYPE | DESCRIPTION & BUSINESS PURPOSE | SOURCE VALUE FROM MOBILE APP | SAMPLE VALUE |
|---|---|---|---|---|
| **Other2** | NVARCHAR(150) | Collector identification and full name tag. | 'Collector: ' + CollectorName | 'Collector: kandy' |
| **PayType** | NVARCHAR(20) | Payment instrument method. | 'CASH' or 'CHEQUE' | 'CASH' / 'CHEQUE' |
| **Bank** | NVARCHAR(100) | Drawee Bank name (for Cheque payments). | Bank name, or empty string "" for Cash | 'Commercial Bank' / '' |
| **ChequeBranch** | NVARCHAR(100) | Bank branch name for cheques, or location fallback for Cash. | Branch name or location string | 'Peradeniya' |
| **ChequeNo** | NVARCHAR(50) | Cheque leaf number (for Cheque payments). | Cheque number, or empty string "" for Cash | 'CHQ994821' / '' |
| **ChequeDate** | DATETIME | Realization date written on cheque (Post-dated or current). | Cheque date, or NULL if Cash | 2026-11-25 / NULL |
| **Details1** | NVARCHAR(250) | Collector field comments and special payment remarks. | Agent notes, or empty string "" | 'Settled before delivery' |
| **Details2** | NVARCHAR(250) | **GPS Coordinates and Physical Geolocation Address**. | Resolved GPS coordinate & physical address | 'GPS: 6.9271, 79.8612 - Colombo' |
| **Details3** | NVARCHAR(150) | Reserved extension attribute. | Empty string default ("") | '' |

## 3. ACCOMPANYING TABLE: DBO.DPAY (DAILY PAYMENT REGISTER)

Simultaneously with AppInvoice, each payment row is inserted into dbo.DPay to feed standard ERP daily collection, cashier reconciliation, and audit reports.

| COLUMN NAME | DATA TYPE | DESCRIPTION & VALUE MAPPING | SAMPLE VALUE |
|---|---|---|---|
| **RNo** | INT | Sequential register payment identifier: ISNULL(MAX(RNo), 0) + 1. | 14820 |
| **Date** | DATETIME | Transaction date (DateTime.Now.Date). | 2026-11-11 |
| **InvNo** | INT | Invoice number (or 0 for Advance payment). | 260928001 |
| **CCode / Name** | INT / NVARCHAR | Customer Code and Customer Business Name. | 65714304 / 'B.I.B. Pharmaceuticals' |
| **Branch** | NVARCHAR(50) | Branch identifier: 'Jay Pharma'. | 'Jay Pharma' |
| **Details** | NVARCHAR(150) | Audit log summary: 'Field Payment by ' + CollectorName. | 'Field Payment by kandy' |
| **Amount** | DECIMAL(18,2) | Settled amount (Rs.) allocated to this record. | 1000.00 |
| **Type** | NVARCHAR(20) | Payment mode: 'Cash' or 'Cheque'. | 'Cash' / 'Cheque' |
| **CheNo / PDDate** | NVARCHAR / NVARCHAR | Cheque number and Post-dated string (format: 'dd/MM/yyyy'). | 'CHQ994821' / '25/11/2026' |
| **[Check] / Audit** | NVARCHAR(50) | Initial verification flags: 'Pending' (ready for head-office accounts clearance). | 'Pending', 'Pending' |
| **MRNo** | INT | Numeric portion of Manual Receipt Number (e.g. from 'KANDY-026482'). | 26482 |

## 4. BUSINESS LOGIC & ALLOCATION SCENARIOS

**Scenario A: Exact or Partial Single Invoice**

Agent collects Rs. 10,000 for Invoice #101 (Val: Rs. 25,000).

- InvNo = 101, InVal = 25000, SetAmt = 10000
- SettleType = 'Part Settled', DOther1 = 15000
- Other1 = 'MR-001', DOther2 = 10000

**Scenario B: Multi-Invoice Split Payment**

Agent collects Rs. 30,000 to settle Invoice #101 (Rs. 10,000) and Invoice #102 (Rs. 20,000).

- **Row 1:** InvNo = 101, SetAmt = 10000, Other1 = 'MR-002'
- **Row 2:** InvNo = 102, SetAmt = 20000, Other1 = 'MR-002'

*Both rows share the exact same Receipt No (Other1).*

**Scenario C: Pure Advance Payment (No Invoices)**

Customer pays Rs. 50,000 on account before billing.

- InvNo = 0, InVal = 0, SetAmt = 50000
- SettleType = 'Advance', DOther1 = 0, DOther2 = 50000
- DPay entry records InvNo = 0 with full amount.

**Scenario D: Overpayment with Excess Balance**

Customer has Rs. 10,000 invoice, but hands over Rs. 15,000.

- **Row 1:** InvNo = 101, SetAmt = 10000, SettleType = 'Settled'
- **Row 2:** InvNo = 0, SetAmt = 5000, SettleType = 'Advance'

*Excess is automatically preserved as an Advance row!*

## 5. HOW ERP COMPUTES EFFECTIVE REALTIME BALANCE (QUERY EXAMPLE)

```
-- Recommended SQL View / Query to compute Real-Time Customer Outstanding: SELECT i.InvNo, i.CCode, i.CName, i.Date AS InvoiceDate, i.InVal AS OriginalAmount, (ISNULL(i.SetAmt, 0) + ISNULL(app.AppSettled, 0)) AS TotalSettledAmount, (i.InVal - (ISNULL(i.SetAmt, 0) + ISNULL(app.AppSettled, 0))) AS EffectiveOutstandingBalance, CASE WHEN (i.InVal - (ISNULL(i.SetAmt, 0) + ISNULL(app.AppSettled, 0))) <= 0.01 THEN 'Settled' WHEN (ISNULL(i.SetAmt, 0) + ISNULL(app.AppSettled, 0)) > 0 THEN 'Part Settled' ELSE ISNULL(i.SettleType, 'Pending') END AS CurrentStatus FROM dbo.Invoice i LEFT JOIN ( SELECT InvNo, SUM(ISNULL(SetAmt, 0)) AS AppSettled FROM dbo.AppInvoice WHERE InvNo > 0 GROUP BY InvNo ) app ON i.InvNo = app.InvNo WHERE (i.InVal - (ISNULL(i.SetAmt, 0) + ISNULL(app.AppSettled, 0))) > 0.01;
```